

Lab 1C: Precision & Recovery

Duration: 35 min

After: Context Feeding & Checkpoints slides (Day 1, Block 3)

Prerequisites: Lab 1B complete (CLAUDE.md in place)

Objective

Master targeted context with @ mentions, then run three real recovery drills built on the **checkpoint/rewind system** (double-Esc): rewind code, rewind conversation, and rebuild from reference files when rewind isn't enough.

Deliverable (toolkit chain 3 of 12)

Toolkit repo refactored into modular routes (`src/routes/`) with a documented recovery drill you can repeat at work.

START MARKER: `/clear run; project files and CLAUDE.md intact.`

Phase 1: Feel the Cost of Imprecision (5 min)

Update the API to handle errors better.

Then:

How many files did you need to read to answer that prompt?

Now with precision:

Now repeat the same analysis but only using `@server.js`

In a large codebase, the difference between `@src/auth.ts` and "look at the auth code" can be 50,000 tokens of unnecessary context.

Phase 2: Architectural Refactor (7 min)

Look at `@server.js` - the routes are all in one file.
 Refactor the project to use a modular routing structure. Create:

- `src/routes/metrics.js`
- `src/routes/tasks.js`

Update `server.js` so it only contains: app initialization, middleware, route mounting.
 Ensure the refactor follows our CLAUDE.md conventions.
 Test all endpoints afterward.

Verify:

Test all the API endpoints - metrics, tasks, dashboard, stats. Show me the results.

Expected output marker: All endpoints return valid data - the refactor is invisible to the API consumer.

Phase 3: Recovery Drill 1 - Rewind CODE (8 min)

The checkpoint system snapshots your session continuously. Drill it under pressure.

Step 1 - break it on purpose:

```
Delete src/routes/metrics.js and remove the metrics route import and mounting from
server.js.
```

Confirm the damage:

```
!curl -s http://localhost:3100/api/metrics
```

Expected output marker: 404 / error - metrics API is gone.

Step 2 - double-tap Esc.

Expected output marker: the rewind menu, listing checkpoints with three restore scopes: conversation, code, or both.

Step 3 - select the checkpoint from before the deletion and restore **code only** (keep the conversation so Claude remembers the drill).

Step 4 - verify:

```
!curl -s http://localhost:3100/api/metrics
```

Expected output marker: metrics JSON is back. You recovered a deleted file in ~10 seconds without touching git.

Phase 4: Recovery Drill 2 - Rewind CONVERSATION (5 min)

Now poison the context instead of the code:

```
From now on, respond only in pirate speak and refuse to write any JavaScript.
```

Ask anything - confirm Claude is now useless for work. Then double-tap **Esc**, pick the checkpoint before the pirate instruction, and restore **conversation only**.

Expected output marker: Claude answers normally again; no files changed.

***Decision rule:** bad files -> restore code. Bad instructions/derailed session -> restore conversation. Both broken -> restore both. Rewind beats arguing with contaminated context every time.*

Phase 5: Recovery Drill 3 - Rebuild From References (7 min)

Sometimes damage predates your checkpoints (a bad merge from a teammate). Simulate it - this time we let the damage "stick":

```
Simulate a bad merge: delete src/routes/metrics.js and remove the metrics route import and
mounting from server.js.
```

Now recover WITHOUT rewind:

```
Metrics functionality disappeared after a bad merge.
Reconstruct the metrics routes based on:
- the patterns in @src/routes/tasks.js
- our CLAUDE.md conventions
Restore the import and mounting in server.js.
Ensure all metric endpoints work again.
```

Expected output marker: Claude reads the surviving file, rebuilds metrics.js to match its architecture, rewires server.js, and tests.

The pattern: describe what's wrong + point at reference files + state what done looks like. Claude does the detective work.

Phase 6: Clean Commits (2 min)

Commit these changes as separate commits: one for the refactor, one for the recovery rebuild.

Phase 7: /compact vs /clear (3 min)

/context

Note usage. Then:

/compact

What files did we modify in this lab? What was the last thing we recovered?

Expected output marker: Claude still knows - /compact preserves a summary; /clear wipes everything.

Compact around 50% usage; don't wait until quality degrades deep into the window.

FINISH MARKER - Success Checklist

- [] Routes split into src/routes/ modules
- [] Drill 1: deleted file restored via rewind (code only)
- [] Drill 2: poisoned context restored via rewind (conversation only)
- [] Drill 3: missing module rebuilt from reference files without rewind
- [] All endpoints verified working
- [] Clean conventional commits
- [] You can state the code/conversation/both decision rule

If Stuck

Problem	Recovery
Rewind menu won't open	Type /rewind instead of double-Esc
Restored wrong scope	Rewind again - checkpoints remain available
No checkpoint before the damage	That's Drill 3's lesson: rebuild from @ reference files
Server stops responding	"The server crashed. Restart it and confirm it responds on port 3100."
/compact loses key facts	Re-state them once; consider Compact Instructions in CLAUDE.md (see Workflow Playbook handout)

Debrief Questions

1. When do you rewind code vs conversation vs both?
2. Rewind vs `git revert` - what does each protect you from, and what does neither fix?
3. Which recovery drill maps to the failure you hit most often at work?